Heat, Temperature, and the TEC Model

Supplement to the Phys 39 oral-exam guide | Module 2 through Module 6 Part I

This guide explains the equation photo step by step, then connects heat, temperature, thermal capacitance, energy units, the TEC heat balance, and the Module 6 oral questions.

1. Heat, temperature, and energy are different

Temperature T describes the thermal state of a material: how hot or cold it is. It is measured in degrees Celsius (°C) or kelvin (K). Temperature is not an amount of energy.

Thermal energy U is energy stored microscopically in the material. Its unit is the joule (J). A warmer or larger object often has more thermal energy, but temperature alone does not tell you the total; mass and material matter too.

Heat Q is energy being transferred because of a temperature difference. It is not a substance stored inside the object. Heat is measured in joules (J). In this guide, Q is positive when energy flows into the object.

Heat-flow rate Q-dot tells how quickly energy is being transferred. Its unit is the watt (W), where 1 W = 1 J/s. A heater rated 40 W transfers energy at 40 joules per second while it is delivering that power.

A temperature difference of 1°C has the same size as a difference of 1 K. Absolute temperatures use an offset; temperature changes do not.

Temperature conversion: $T_K = T_{°C} + 273.15$
Difference conversion: ΔT in K = ΔT in °C
Power and energy: 1 W = 1 J/s; 1 Wh = 3600 J; 1 kJ = 1000 J

A common oral-exam answer: “Heat is energy crossing a boundary; temperature is a state variable describing the object. The heat transferred is in joules; temperature is in °C or K.”

2. Thermal capacitance: what the cup example means

Thermal capacitance C_{th} is the energy needed to raise an object’s temperature by one degree. It depends on both the amount of material and the material’s specific heat capacity.

$C_{th} = m\, c_p$
For a well-insulated object: ΔU is about $Q = m\, c_p\, \Delta T = C_{th}\, \Delta T$
C_{th} units: J/K or J/°C; c_p units: J/(kg·K)

Here m is mass in kilograms, c_p is specific heat capacity, and ΔT is the temperature change. The approximation assumes c_p is nearly constant and there is no phase change. During a real experiment, some input energy may also leak to the surroundings, so Q into the heater is not always equal to the object’s increase in stored energy.

Worked example: warming 250 mL of water from 20°C to 30°C

For water, density is about 1.00 kg/L, so 250 mL = 0.250 L has mass m about 0.250 kg. Use c_p about 4184 J/(kg·K).

$C_{th} = m\, c_p = 0.250 \times 4184 = 1046$ J/K
$\Delta T = 30 - 20 = 10°C = 10$ K
Q is about $C_{th}\, \Delta T = 1046 \times 10 = 10{,}460$ J = 10.46 kJ
If this took 300 s: average Q-dot is about 10,460/300 = 34.9 W

What if you put a cup down? The cup, water, and contact surface exchange heat. Heat flows from the warmer parts to the cooler parts, so the coffee may cool while the cup warms. A heavier cup or a material with larger c_p has larger capacitance and needs

more energy per degree. The whole system approaches a shared equilibrium temperature if it is insulated from the room.

For two objects mixed together with no heat lost to the room, energy lost by the hot object equals energy gained by the cold one:

$$C_{hot}(T_{hot} - T_f) = C_{cold}(T_f - T_{cold})$$
$$T_f = [C_{hot}T_{hot} + C_{cold}T_{cold}] / [C_{hot} + C_{cold}]$$

3. The conversion constants to know

Quantity	Conversion	Meaning
Celsius to kelvin	T(K) = T(°C) + 273.15	Use for absolute temperature in thermistor equations.
Kelvin to Celsius	T(°C) = T(K) − 273.15	Subtract the offset.
Temperature interval	1 K change = 1°C change	No 273.15 offset for ΔT.
Water specific heat	c_p about 4184 J/(kg·K)	Useful approximate value near room temperature.
Power	1 W = 1 J/s	Energy transfer per second.
Watt-hour	1 Wh = 3600 J	One watt sustained for one hour.
Kilojoule	1 kJ = 1000 J	A convenient energy scale.

Important notation trap: C_{th} means thermal capacitance and has units J/K. °C means degrees Celsius. They are different things despite both using the letter C.

4. Your whiteboard photo: the P-controlled heat balance

The photo begins with a one-lump energy balance, writes the P controller, substitutes it, sets the derivative to zero, groups terms, and solves for $T_{__}$. Read each line as one logical step.

First, a dimensional correction: if P_u is the TEC power coefficient in W per PWM count, the full TEC power term is $P_u u$. The first line in the photo appears to omit u; the later substitution correctly includes the commanded PWM.

Line 1 — energy in minus energy out equals energy storage rate:

$$C_{th}\, dT/dt = P_u u - H(T - T_{amb})$$

Left side $C_{th}dT/dt$ is in watts: (J/K)(K/s) = J/s. $P_u u$ is TEC power: (W/count)(counts) = W. $H(T-T_{amb})$ is passive heat-loss rate: (W/K)(K) = W. All terms are powers, so the equation is dimensionally consistent.

Line 2 — define the P-control command. The error is target minus measured temperature:

$$e = T_{set} - T$$
$$u = K_p e = K_p(T_{set} - T)$$

K_p units are PWM counts per degree. Multiplying by error in degrees gives PWM counts. A positive error requests heat; a negative error requests cooling.

Line 3 — substitute u into the heat balance:

$$C_{th}\, dT/dt = P_u K_p(T_{set} - T) - H(T - T_{amb})$$

The first term now says how much TEC power the controller asks for. The second term says how fast heat leaves toward ambient.

Line 4 — impose steady state. Steady means the temperature is no longer changing, so dT/dt = 0. The individual heat flows need not be zero; their net is zero:

$$0 = P_u K_p (T_{set} - T) - H(T - T_{amb})$$

Line 5 — expand both parentheses. Distribute each multiplier:

$$0 = P_u K_p T_{set} - P_u K_p T - HT + HT_{amb}$$

Line 6 — collect terms containing T on the left. Add $P_u K_p T$ and HT to both sides:

$$(P_u K_p + H)T = P_u K_p T_{set} + HT_{amb}$$

Line 7 — divide by the coefficient multiplying T. This gives the equilibrium temperature:

$$T_{ss} = [P_u K_p T_{set} + HT_{amb}] / [P_u K_p + H]$$

Why is it below the setpoint? At T_{set}, the P error would be zero, so P would request zero power. If the plate is above ambient, it would then lose heat. A small remaining error is needed to command the heat that balances the loss. That remaining difference is called droop.

5. Simplify the photo's result with measured parameters

Divide the steady-state equation through by H and define the heating susceptibility chi = P_u/H. Then $L = K_p chi$ is dimensionless:

$$T_{ss} = [L\,T_{set} + T_{amb}] / (1 + L)$$
$$\text{Droop} = T_{set} - T_{ss}$$
$$= [T_{set}(1 + L) - (L\,T_{set} + T_{amb})] / (1 + L)$$
$$= [T_{set} + L\,T_{set} - L\,T_{set} - T_{amb}] / (1 + L)$$
$$= (T_{set} - T_{amb}) / (1 + L)$$

The L*Tset terms cancel because the same proportional term appears in the equilibrium expression and in the target being subtracted. The result shows that greater loop gain leaves a smaller fraction of the initial error.

Worked Module 5 prediction: Tset = 30.00°C, Tamb = 21.54°C, chi_h = 0.50127°C/count, Kp = 2.00 counts/°C.

$$\text{Initial error } e_0 = 30.00 - 21.54 = 8.46°C$$
$$L = K_p chi = 2.00 \times 0.50127 = 1.00254$$
$$\text{Droop} = e_0/(1 + L) = 8.46/2.00254 = 4.2246°C$$
$$T_{ss} = 30.00 - 4.2246 = 25.7754°C, \text{ about } 25.78°C$$

The earlier Module 5 log measured about 25.75°C at Kp = 2, a difference of roughly 0.03°C from this prediction. Ambient temperature was not logged throughout the sweep, so the small difference is not proof the model is exact.

6. Module 6 Part I oral questions

What physical parts store heat? The aluminum plate/block, TEC ceramic faces, thermistor, exchanger, mounting hardware, and any attached fluid. The one-lump model treats them as one effective temperature and one total capacitance.

What paths let heat leave the measured block? Conduction through the TEC to the reservoir and exchanger, conduction through mounts/wires, convection to room air, and some radiation. The one-lump H combines these paths.

What evidence suggests a time constant? After a fixed PWM step, temperature starts changing quickly and then approaches a final value more slowly. Module 4 traces fit exponentials with tau about 62.9 ± 5.0 s.

Why does the algebraic droop model not predict oscillation? It sets dT/dt = 0, so it predicts only the final equilibrium, not the path to it.

Why can the dynamic one-lump P model not oscillate? It has one state and one real negative eigenvalue, giving one monotonic exponential. It cannot overshoot or cross equilibrium.

What happens if thermal capacitance doubles? For the same net power, dT/dt is halved and tau = C/H doubles. The steady-state temperature does not change.

What happens if the object is twice the mass of the same material? Its Cth = m cp doubles. Twice as much energy is needed for the same temperature change, absent losses or phase change.

What happens if H doubles? Passive heat flow per degree doubles; chi = Pu/H halves and tau = C/H halves if C stays fixed.

What happens if PWM is zero? TEC drive power is zero, but passive heat flow continues. A warmer plate cools toward room; a colder plate warms toward room.

What does a watt measure? Power, or energy-transfer rate: one joule per second. It is not an amount of energy.

Quick unit-check for the TEC balance

Term	Units	Interpretation
Cth dT/dt	(J/K)(K/s) = W	rate thermal energy changes
Pu u	(W/count)(counts) = W	TEC power into/out of lump
H(T − Tamb)	(W/K)(K) = W	passive heat-loss rate
m cp ΔT	kg · J/(kg K) · K = J	energy needed for temperature change
τ = C/H	(J/K)/(W/K) = s	characteristic response time

Source scope: equations from the supplied Module 6 Part I page and your photographed whiteboard derivation; numerical examples from the saved Module 2 and Module 5 notes. This is a separate supplement and does not replace your main oral-exam guide.